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Vertical power devices having mesas and etched trenches therebetween

Abstract

A vertical semiconductor and method for fabricating the same is disclosed. In one embodiment, fabrication entails providing a precursor comprising a substrate and a drift region over the substrate. A plurality of trenches is etched into the drift region from a top surface of the drift region such that a plurality of mesas remains in an upper portion of the drift region. The plurality of trenches is then filled with a first material. A vertical semiconductor device includes a plurality of mesas extends from an upper portion of the drift region, wherein there are no regrowth interfaces between the drift region and the plurality of mesas. A first material fills the trenches between each one of the plurality of mesas. At least one first contact over at least one of the plurality of mesas. At least one second contact over a bottom surface of the substrate.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to vertical field effect devices with improved performance and efficient fabrication techniques for making the same.

BACKGROUND

(2) Vertical semiconductor devices come in several varieties, each of which may be used for different applications. One notable use of vertical semiconductor devices is for high power

applications. In particular, devices such as PiN diodes, Schottky diodes, metal-oxide semiconductor field-effect transistors (MOSFETs), junction field-effect transistors (JFETs), and the like may be rated for high blocking voltages and on-state currents, and thus are often used for these power applications. Given the ever-increasing pressure to provide higher performing devices at lower costs, there is continuing need for novel device structures and fabrication techniques that result in higher performance, lower cost, or both.

SUMMARY

(3) A vertical semiconductor and method for fabricating the same is disclosed. In one embodiment, fabrication entails initially providing a precursor comprising a substrate and a drift region over the substrate, wherein the substrate and the drift region are doped with a dopant of a first type and formed from a first material system. A plurality of trenches is etched into the drift region from a top surface of the drift region such that a plurality of mesas remains in an upper portion of the drift region. The plurality of trenches is then filled with a first material doped with a dopant of a second type, which is opposite in polarity as the first type. At least one first contact is provided over at least one of the plurality of mesas, and at least one second contact is provided over a bottom surface of the substrate.

(4) In another embodiment, a vertical semiconductor device includes a substrate and a drift region over the substrate. The substrate and the drift region are doped with a dopant of a first type and formed from a first material system. A plurality of mesas extends from an upper portion of the drift region, wherein there are no regrowth interfaces between the drift region and the plurality of mesas. A first material doped with a dopant of a second type fills the trenches between each one of the plurality of mesas, and the second type is opposite in polarity to the first type. At least one first contact is formed over at least one of the plurality of mesas. At least one second contact is formed over a bottom surface of the substrate.

(5) The first material system may be silicon carbide and the first material may be silicon carbide, silicon, or a metal. During fabrication, the step of filling the plurality of trenches may include providing a first layer of the first material over the top surface of the drift region such that the plurality of trenches is filled and the plurality of mesas are covered, and planarizing the first layer to expose the plurality of mesas.

(6) The vertical semiconductor device may be a transistor or diode. For a transistor, remaining portions of the first layer that fill the plurality of trenches form a plurality of gate regions, the at least one first contact is a source contact, and the at least one second contact is a drain contact. At least one gate contact is provided over at least a portion of the gate layer after planarizing the first layer.

(7) In some embodiments, the at least one gate contact is not provided over the at least one gate region, wherein in others, the at least one gate contact includes a plurality of gate contacts that are each provided over a corresponding one of the plurality of gate regions.

(8) In certain embodiments, after planarizing the first layer, a second layer is formed from the first material system over the plurality of mesas and remaining portions of the first layer, wherein the at least one first contact is provided over a portion of the first layer.

(9) The drift region and the plurality of mesas may be continuously grown such that there are no regrowth interfaces between the drift region and the plurality of mesas. Further, the dopant of the first type may be an N-type dopant, and the dopant of the second type may be a P-type dopant in certain embodiments and the opposite in other embodiments.

(10) In one embodiment, each of the plurality of mesas are elongated stripes such that one of the plurality of trenches is provided between adjacent pairs of the plurality of mesas. Alternatively, each of the plurality of mesas may be pillars having horizontal cross-section of virtually any shape such as circular, square, rectangular, polygonal, or the like.

(11) The drift region may have a plurality of regions with different doping concentrations. Each of the plurality of mesas may have a plurality of regions with different doping concentrations. One or

more of the plurality of mesas or the drift regions may have at least one region with a graded doping profile. The drift region may have at least one charge spreading layer below the plurality of mesas and at least one drift layer.

(12) In one embodiment, a doping concentration of the second type for the first material is at least two times that of a doping concentration of the first type in the plurality of mesas.

(13) In one embodiment, a width at a most narrow portion of each of the plurality of mesas is between one and two times that of a height of each of the plurality of mesas.

(14) Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure. The FIGS. identify exemplary doping polarities, or types, (N-type and P-type) for the various layers, regions, and/or sections of the vertical semiconductor devices for concise and readability. These polarities, or types, may be reversed in alternative embodiments.

(2) FIGS. 1A and 1B are a cross-sectional view and a top view of a junction field-effect transistor (JFET) according to a first related art.

(3) FIG. 2 is a sectional view of a transistor according to a second related art.

(4) FIGS. 3A-3F illustrate part of a first exemplary fabrication process according to one embodiment of the present disclosure.

(5) FIGS. 4A-4C illustrate part of a second exemplary fabrication process according to one embodiment of the present disclosure.

(6) FIG. 5 is a cross-section of a JFET according to one embodiment of the present disclosure.

(7) FIG. 6 is a JFET according to one embodiment of the present disclosure.

(8) FIGS. 7A-7K illustrate part of a first exemplary fabrication process according to one embodiment of the present disclosure.

(9) FIGS. 8A-8C illustrate part of a first exemplary fabrication process according to one embodiment of the present disclosure.

(10) FIGS. 9A-9K illustrate part of a first exemplary fabrication process according to one embodiment of the present disclosure.

(11) FIG. 10 is a cross-section of a JFET according to one embodiment of the present disclosure.

(12) FIG. 11 is a cross-section of a JFET according to one embodiment of the present disclosure.

(13) FIG. 12 is a cross-section of a vertical diode according to one embodiment of the present disclosure.

(14) FIG. 13 is a cross-section of a vertical diode according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(15) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description considering the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. These concepts and applications fall within the scope of the disclosure and the accompanying claims.

(16) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to

distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes all combinations of one or more of the associated listed items.

(17) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

(18) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

(19) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(20) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Doping concentrations are referenced using the nomenclature: $aE_b \text{ cm.sup.-3}$, which represents $a \times 10^b \text{ cm.sup.-3}$ and where cm is centimeters. For example, $5E16 \text{ cm.sup.-3}$ equals $5 \times 10^{16} \text{ cm.sup.-3}$, where a is 5 and b is 16. The Figures identify exemplary doping polarities, or types, (N-type and P-type) for the various layers, regions, and/or sections of the vertical semiconductor devices for concise and readability. These polarities, or types, may be reversed in alternative embodiments.

(21) The present disclosure relates to vertical semiconductor devices, such as transistors, diodes, and the like as well as fabrication techniques for building the same. In one embodiment, fabrication entails initially providing a precursor comprising a substrate and a drift region over the substrate, wherein the substrate and the drift region are doped with a dopant of a first type and formed from a first material system. A plurality of trenches is etched into the drift region from a top surface of the drift region such that a plurality of mesas remains in an upper portion of the drift region. The plurality of trenches is then filled with a first material doped with a dopant of a second type, which is opposite in polarity as the first type. At least one first contact is provided over at least one of the plurality of mesas, and at least one second contact is provided over a bottom surface of the substrate.

(22) In another embodiment, a vertical semiconductor device includes a substrate and a drift region over the substrate. The substrate and the drift region are doped with a dopant of a first type and formed from a first material system. A plurality of mesas extends from an upper portion of the drift region, wherein there are no regrowth interfaces between the drift region and the plurality of mesas. A first material doped with a dopant of a second type fills the trenches between each one of the plurality of mesas, and the second type opposite in polarity as the first type. At least one first contact is formed over at least one of the plurality of mesas. At least one second contact is formed over a bottom surface of the substrate. The first material system may be silicon carbide and the first material may be silicon carbide, silicon, or a metal.

(23) Details for these fabrication techniques and device structures are provided below. Specific examples of junction field-effect transistors (JFETs) are illustrated; however, the concepts provided apply to other implanted regions of other types of transistors, such as MOSFETs and IGBTs as well as to junction barrier Schottky (JBS), merged PiN Schottky (MPS), and like diodes by modifying the contact structure and placement, as those skilled in the art will appreciate. The associated figures for the text below relate to a striped device structure of conducting channels that extend into the plane of the drawings, but the layouts are not limited to such structure. The surface layout could be stripes, rectangles, circles, polygons, and the like.

(24) FIG. 1A illustrates a cross-sectional view of a junction field-effect transistor (JFET) **10A** of the related art. The JFET **10A** is a vertical JFET in that current flows vertically through the device between the source (S) and the drain (D) based on the voltage applied at the gate (G). In the illustrated embodiment, the JFET **10A** includes a substrate **12**, a drift region **14** over the substrate **12**, and a channel region **16** over the drift region **14**. The upper portion of the body of the JFET **10A** is etched to form a mesa **18**. Gate regions **20** are formed in opposing sides of the mesa **18** and extend downward and outward along the upper portion of the body of the JFET **10A**. As such, the channel region **16** resides between the gate regions **20** in the mesa **18** as well as between the bottom of the respective gate regions **20** and the top of the drift region **14**.

(25) An ohmic region **22** is formed in the top of the mesa **18**, wherein a source contact **24** is provided on the top of the ohmic region **22**. Gate contacts **26** are formed on the recessed portions that reside on either side of the mesa **18**. The gate contacts **26** reside over the horizontal sections of the respective gate regions **20**. A drain contact **28** resides along the bottom of the substrate **12**.

(26) As illustrated, the body of the JFET **10A** is silicon carbide (SiC); however, other semiconductor material systems, such as silicon (Si), Germanium (Ge), Silicon Germanium (SiGe), Gallium Nitride (GaN), Aluminum Nitride (AlN), Gallium Oxide (Ga.sub.2O.sub.3), and diamond are applicable. The substrate **12**, drift region **14**, channel region **16**, and gate regions **20** may be SiC, wherein each section is formed by implanting N-type or P-type dopants at varying locations and concentrations within the body of the JFET **10A**. In the illustrated example, the substrate **12** is moderately doped with an N-type dopant, the drift region **14** is moderately doped with an N-type dopant, the channel region **16** is moderately doped with an N-type dopant, the gate regions **20** are heavily doped with a P-type dopant, and the ohmic region **22** is heavily doped with an N-type dopant. As those skilled in the art will appreciate, the actual and relative doping levels for each region of the JFET **10A** will depend on the desired characteristics of the JFET **10A** and may vary from one application to another. The polarity of the doping for each region may be reversed from what is illustrated in the embodiments that follow. The source, gate, and drain contacts **24**, **26**, **28** may be formed from an appropriate metal or other highly conductive material.

(27) The vertical portion of the mesa **18** that resides between the gate regions **20** provides the JFET conducting gap CG. The illustrated section is a cross-section of a full repeating cell in a larger device that comprises many cells, which are striped in the Z-direction (into the figure). The mesa **18** is formed by an etching process, and the gate regions **20** are formed using angled implants at different orientations. In the illustrated example, the implants are provided at four different angles. As such, fabrication of this embodiment is a complex process, and the more complex the process,

the more expensive it is to fabricate the device. To further complicate matters, the amount of the P-type dopant along the sides of the mesa **18** necessary to form the gate regions **20** is a function of the angle of the sidewalls of the mesa **18**. Angled implants at four incident directions are made to achieve P+ type gate regions **20**, which generally surround each stripe of the N-type channel region **16** that is within the mesa **18**. As such, etch control for the formation of the mesa **18** is a critical, complex, and relatively expensive process.

(28) FIG. **1B** is a cross-sectional SEM image of the device showing the gate contacts **26** and source fingers **S** running in the z-direction. Further information can be found in Kiran Chatty, David C Sheridan, Volodymyr Bondarenko, Robin Schrader, and Jeffrey B Casady, "900V, 1.46 mOhm-cm² 4H—SiC Depletion Mode Vertical JFET," Materials Science Forum, Vols. 740-742, pp 915-920 (2013), which is incorporated by reference in its entirety.

(29) FIG. **2** illustrates another JFET structure **10B** of the related art that includes P+ type gate stripes **30** that are buried in an N-type drift region **32**. The drift region **32** resides over an N+ type SiC substrate **34**, which includes a drain contact **36** on a bottom portion thereof. The gate stripes **30** are created by forming a P-type layer (not shown) and etching that layer to leave the gate stripes **30** that are illustrated. Subsequently, N-type material is used to fill in the trenches between the gate stripes **30** to provide a layer of N-type material over the gate stripes **30** to provide the remainder (i.e. upper portion) of the drift region **32** such that the gate stripes **30** are buried in the drift region **32**. The upper portions of the drift region **32** that is between the gate stripes **30** are the channel regions. A disadvantage of this type of process is that the channel regions and the drift region **32**, which carry the current for the JFET **10B** are not continuously grown. Whenever current must flow through an interface where the growth process has been stopped and then started again, inefficiencies are induced due to the breaks in the lattice structure at the interface between the two regions.

(30) An ohmic contact **38** is formed in an upper portion of the layer where the source contact **40** resides. A gate contact **42** is connected to a gate area **30A** that is formed with the gate stripes **30**, so that every gate area **30A** or gate stripe is connected. The long gate stripes lead to high resistance and slow the switching speeds, and a portion of the drift region **32** is formed by epi regrowth after etching. Unfortunately, the drift region **32** is the critical semiconductor portion that carries all the device current and should be as defect free as possible to optimize performance. Further information can be found in Yasunori Tanaka, Koji Yano, Mitsuo Okamoto, Akio Takatsuka, Kazuo Arai and Tsutomu Yatsuo, "1270V, 1.21 mΩ.Math.cm² SiC Buried Gate Static Induction Transistors (SiC-BGSITs)," Materials Science Forum, Vols. 600-603, pp 1071-1074 (2009), which is incorporated herein by reference in its entirety. As provided in the embodiments below, improvements over the related art are provided by having the channel regions formed in an original epitaxial layer to avoid the defects inherent in regrowth processes.

(31) FIGS. **3A** through **3F** illustrate a first exemplary process for fabricating a vertical semiconductor device according to one embodiment of the disclosure. Initially, a precursor **50** is provided as illustrated in FIG. **3A**. The precursor **50** may be continuously grown SiC or the like having a substrate **52**, which is heavily doped (N+) with an N-type dopant, a drift region **54** over the substrate, which is moderately doped (N) with an N-type dopant, and an ohmic layer **56** over the drift region **54**, which is heavily doped (N+) with an N-type dopant. The ohmic layer **56** may be continuously grown with the precursor **50** as noted above or grown as a separate epitaxial layer.

(32) Next, trenches **58** are etched into the top surface of the precursor **50**, wherein the trenches **58** are in locations where gate regions **64** (FIGS. **3D** and **3E**) will be formed in subsequent steps, as illustrated in FIG. **3B**. The unetched portions of the precursor **50** between the trenches **58** are referred to as mesas **60**. Each mesa **60** provides a channel region **62** and an ohmic region **64** over the channel region **62**. The ohmic region **64** for each mesa **60** is formed from an unetched portion of the ohmic layer **56** while each channel region **62** is formed from an unetched portion of the drift region **54**.

(33) Once the trenches **58** are formed, a gate layer **66** is formed in the trenches **58** and over the top portions of the ohmic regions **64** of the mesas **60**, as illustrated in FIG. 3C. The gate layer **66** is doped with a P-type dopant. In the illustrated embodiment, the gate layer **66** is P-type silicon carbide; however, P-type silicon or an appropriate p-type conductor, such as nickel oxide, may also be used. For P-type silicon, use of an epitaxial film is not necessary. The trenches **58** may be simply filled with the P-type silicon by chemical vapor deposition or other similar film deposition methods. Other materials for the gate layer and/or the gate regions **68** include any other P-type material that is chemically compatible with SiC. At this point, the gate layer **66** is then planarized, if necessary, using an etch process or the like to remove the upper portion of the gate layer **66** that extends above the top surface of the mesas **60** such that the ohmic regions **64** are exposed, as illustrated in FIG. 3D. The remaining portions of the gate layer **66** that fill the trenches **58** provide gate regions **68**.

(34) After planarizing, gate contacts **70** are formed on top of each gate region **68**, and source contacts **72** are formed on the ohmic regions **64** over the channel regions **62** for each mesa **60**, as illustrated in FIG. 3E. A drain contact **74** is formed on the bottom of the substrate **52** at this point, if the drain contact **74** was not already provided on the precursor **50**. In one embodiment, the mesas **60**, channel regions **62**, gate regions **68**, source contacts **72**, and the gate contacts **70** are linear stripes that extend into FIG. 3E. These features may take on a variety of shapes, as will be described further below.

(35) An alternative process for forming the source contacts **72** is provided in FIG. 3F. The process through FIG. 3D remains the same as that described above. As such, the gate contacts **70** are formed on top of each gate region **68**, but the source contacts **72** are not formed as stripes at this stage. Instead, a dielectric **83**, such as an appropriate gate oxide **83**, is formed over the gate contacts **70** such that the ohmic regions **64** over the channel regions **62** remain exposed. A metal or other conductive material is then formed over the top of the precursor **50** to form a unified source contact **72'**, which covers the dielectric **83** and the ohmic regions **64** such that the unified source contact **72'** makes electrical contact with the ohmic regions **64** and is electrically isolated from the gate contacts **70**.

(36) For these embodiments, there is no need for an angled implant as required for the JFET **10A** of FIGS. 1A and 1B, and a trench etch is used in the N-type SiC of the precursor **50** instead of the P-type layer as done for the JFET of FIG. 2. As such, current is carried through the continuously grown N-type epitaxial channel regions **62** and drift region **54** instead of a regrown epitaxial structure as required for the JFET **10B** of FIG. 2.

(37) In select embodiments, the substrate **52** is heavily doped with an N-type dopant at exemplary concentrations in the range of $1\text{E}18$ to $1\text{E}19$ cm.sup.-3; $1\text{E}19$ to $1\text{E}20$ cm.sup.-3; or $1\text{E}18$ to $1\text{E}19$ cm.sup.-3, and the drift region **54** is moderately doped with an N-type dopant at exemplary concentrations in the range of $1\text{E}15$ to $1\text{E}16$ cm.sup.-3; $1\text{E}16$ to $2\text{E}17$ cm.sup.-3; or $1\text{E}15$ to $2\text{E}17$ cm.sup.-3, the channel regions **62** are moderately doped with an N-type dopant at exemplary concentrations in the range of $1\text{E}15$ to $1\text{E}16$ cm.sup.-3; $1\text{E}16$ to $2\text{E}17$ cm.sup.-3; or $1\text{E}15$ to $2\text{E}17$ cm.sup.-3, the gate regions **68** are heavily doped with a P-type dopant at exemplary concentrations in the range of $5\text{E}16$ to $1\text{E}18$ cm.sup.-3; $1\text{E}18$ to $1\text{E}20$ cm.sup.-3, or $5\text{E}16$ to $1\text{E}20$ cm.sup.-3, the ohmic regions **64** are heavily doped with an N-type dopant at exemplary concentrations in the range of $1\text{E}18$ to $1\text{E}19$ cm.sup.-3; $1\text{E}19$ to $1\text{E}21$ cm.sup.-3; or $1\text{E}18$ to $1\text{E}21$.

(38) In certain embodiments, the doping concentration levels for the gate regions **68** are at least 1.5 times, 2 times, or 2.5 times that of the channel regions **62**, which are provided by the mesas **60** that extend vertically upward from the top of the drift region **54**. Generally, the doping of the gate regions **68** should be high enough to allow the channel regions **62** to deplete under reverse bias conditions. The height of the mesas **60** (channel regions **62**) may be one to two times ($1\times$ - $2\times$) that of the width of the mesas **60** (channel regions **62**). The width of the trenches **58** may range from 0.5 microns to 5 microns, 0.75 micron to 4 microns, 1 micron to 3.5 microns, 1 micron to 4 microns,

and 1.5 to 3.5 microns.

(39) The concepts provided are particularly beneficial in power applications. Such applications are defined as those having blocking voltages of more than 400V when reversed biased, capable of passing current more than 1 A when forward biased, and/or having a power rating of at least 2 Watts.

(40) These are merely exemplary doping levels and dimensions, wherein embodiments that incorporate the inventive concepts described herein and within the scope of the claims that follow may fall anywhere within or outside of the ranges listed above depending on the desired performance parameters of the device. Unless otherwise stated, these exemplary doping concentrations apply to the embodiments that follow. Where graded doping levels are discussed, the graded doping profiles may range from the lower to the higher ends of the stated ranges and increase or decrease in concentration from the bottom to the top of the particular region.

(41) Another embodiment is illustrated in FIGS. 4A through 4C. The process is the same as that of FIGS. 3A through 3D, wherein the ohmic regions 64 at the top of the mesas 60 are optional. Once the gate layer 66 is planarized to form the various gate regions 68, an isolation layer 79 is applied over a portion of the top surface of the precursor 50 and an ohmic layer 80 is applied over the top surface of the isolation layer 79 using a deposit and etching process or the like, as illustrated in FIG. 4A. The isolation layer 79 and the ohmic layer 80 covers portions of the gate regions 68 and channel regions 62 where one or more source contacts 82 are to be applied. The one or more source contacts 82 are then formed over the ohmic layer 80, as illustrated in FIG. 4B. Gate contacts 84 may be formed directly over an exposed gate region 68G in this or a separate step, as illustrated in FIG. 4C. The gate contact 84 makes ohmic contact with the P-type gate region 68G, which is an extension of the gate regions 68, but does not make ohmic contact with the source contact 82. In other words, the metals for the respective source contact 82 and gate contact 84 are isolated from each other. The isolation layer 79 and the ohmic layer 80 may be formed as an integral layer with different N-type doping levels or in separate steps.

(42) In this embodiment, a gate connector region 68G is formed perpendicular to and intersects with the various gate regions 68, which are elongated stripes in this embodiment, as illustrated in FIG. 4C. The gate contact 84 resides over the gate connector region 68G, wherein there is no N-type material, such as the ohmic layer 80 between the gate contact 84 and the gate connector region 68G. An upper portion of or an additional layer over the gate connector region 68G may be heavily doped (P-type) to provide an ohmic region to facilitate the electrical connection between the gate contact 84 and the gate connector region 68G. The gate regions 68 may also be epitaxially grown with the rest of the precursor 50 or formed from deposited silicon or other P-type material. This is possible because the P-type gate material does not carry the primary device current.

(43) For any of the embodiments described herein, any of the various regions and layers may have the same or different doping levels using uniform and/or graded doping profiles. FIG. 5 illustrates some of these concepts, which may be provided individually or in combination. The drift region 54 may be divided into multiple layers 88, 90, wherein a charge spreading layer 86 may be provided near an upper portion of the drift region 54 and just below the mesas 60, such that the mesas 60 effectively extend from the charge spreading layer 86. One or more drift layers 88, 90 may be provided below the charge spreading layer 86. Each of these layers may have a uniform or graded doping profile. Further, the channel regions 62 may have a uniform or graded doping profile 62P with any of the doping combinations for the drift region 54. These doping options provide the designer with the ability to tailor the electrical properties of the device, such as the on-state and off-state properties.

(44) FIGS. 6 and 7A-7J illustrate embodiment where the N-type mesas 60 and channel regions 62 take the form of pillars, as opposed to the elongated stripes of the prior embodiments. The pillar-shaped channel regions 62 may a horizontal cross-section of virtually any shape, such as square, rectangular, circular, octagonal, hexagonal, triangular, or the like. This grid-like layout may reduce

gate resistance by reducing contact connection distances. Such embodiments may provide for a larger source contact **82**, as illustrated relative to the previous embodiments, while reducing layout challenges associated with having to work around numerous elongated gate contacts **84**. The N-type isolation layer **79** and ohmic layer **80** effectively isolate the source contact **82** from the gate regions **68**.

(45) FIGS. 7A through 7K illustrate notable steps of an exemplary fabrication process for the embodiment of FIG. 6. FIGS. 7A and 7B are top and side views of the precursor **50** after the etch step. In this embodiment, the mesas **60** have a square cross-section and are aligned in a grid pattern. FIGS. 7C and 7D are top and side views of the precursor **50** after the gate layer **66** (not shown) is planarized to form the gate region(s) **68**. In this embodiment, there is a gate area **68G** on the precursor that has no mesas **60** and is reserved for the gate contact **84** (not shown). Notably, the portion of the gate region(s) **68** in the gate area **68G** is integrally formed and connected with those portions surrounding the mesas **60**. The mesas **60** provide the channel regions **62**, wherein the upper portions of the mesas **60** are heavily doped to provide ohmic contacts **64**.

(46) FIGS. 7E and 7F are top and side views of the precursor **50** after the isolation layer **79** is formed over the active region of the device. FIGS. 7G and 7H are top and side views of the precursor **50** after the N-type ohmic layer **80** is provided over the isolation layer **79** over the isolation layer **79**, such that the N-type isolation layer **79** and the ohmic layer **80** do not cover the gate area **68G**. After the N-type ohmic layer **80** is provided, the source contact **82** is provided over ohmic layer **80**, and the gate contact **84** is provided over the gate area **68G**, as illustrated in FIGS. 7I-7K. The source contact **82** and the gate contact **84** are separated from each other. FIGS. 8A, 8B, and 8C illustrate mesas that have a hexagonal cross-section and correspond to FIGS. 7A, 7C, and 7E.

(47) For a source contact configuration like that of FIG. 3F, reference is made to FIGS. 9A through 9K. FIGS. 9A and 9B are top and side views of the precursor **50** after the etch step. In this embodiment, the mesas **60** have a square cross-section and are aligned in a grid pattern. FIGS. 9C and 9D are top and side views of the precursor **50** after the gate layer **66** (not shown) is planarized to form the gate region(s) **68**. In this embodiment, there is a gate area **68G** on the precursor that has no mesas **60** and is reserved for the gate contact **84** (not shown). Notably, the portion of the gate region(s) **68** in the gate area **68G** is integrally formed and connected with those portions surrounding the mesas **60**. The mesas **60** provide the channel regions **62** wherein the upper portions of the mesas **60** are heavily doped to provide ohmic contacts **64**.

(48) FIGS. 9E and 9F are top and side views of the precursor **50** after the dielectric layer **83'** is formed over the active region of the device. FIGS. 9G and 9H are top and side views of the precursor **50** after openings **85** are formed in the dielectric layer **83** to expose the ohmic regions **64** of the channel regions **62**. The dielectric layer **83** is formed or etched to not cover the gate area **68G**. As illustrated in FIGS. 9I-9K, the source contact **82** is provided over dielectric layer **83** and into the openings **85** to cover the exposed surfaces of the ohmic layer **64**. The gate contact **84** is provided over the gate area **68G**.

(49) Embodiments like those described above may have various doping profiles and additional layers as discussed with FIG. 5. Such an embodiment is illustrated in FIG. 10. Note the presence of the charge spreading layer **86** and drift layers **88**, **90** in the drift region **54**. Each of the layers as well as the mesas **60** or source regions **62** may also have graded doping profiles. FIG. 11 illustrates a variant where the mesas **60**/source regions **62** are elongated with a rectangular cross-section and the different regions are doped as described above in association with FIGS. 5, 6, and 10.

(50) As noted, the concepts provided herein may be applied to various types of vertical semiconductor devices. FIGS. 12 and 13 illustrate exemplary vertical diode structures for vertical diodes **90A** and **90B**, respectively. The fundamental differences between the transistor devices described above and the vertical diodes **90A** and **90B** is the absence of a gate contact and the expansion of the source contact **92** and the underlying active area. The details and structure of the

precursor **50** is essentially the same.

(51) Diode **90A** of FIG. **12** provides a grid of mesas **60** for the source regions **62**, wherein the mesas **60** were formed using an etching process, and the resulting trenches were filled using deposition or growth processes with a P-type material to generate P regions **96** (i.e. gate regions). The trenches and resulting P regions **96** provide a cross-hatched pattern that surrounds a majority of the mesas **60**.

(52) Diode **90B** of FIG. **13** further differs in that the mesas **60** interconnect and form a cross-hatched pattern. The P regions **96** are formed in holes that were etched into the drift region **54** and filled with P-type material. The holes, and thus the P regions **96**, may take on virtually any geometry, such as having a horizontal cross-section that is square, rectangular, round, hexagonal, octagonal, triangular, and the like. In this embodiment, the mesas **60** form a cross-hatched pattern that surrounds the individual P regions, and mostly unconnected, P regions **96**. The doping and additional layers provided in FIGS. **9** and **10** may be applied to the embodiments of FIGS. **12** and **13**.

(53) Those skilled in the art will recognize additional improvements and modifications to the disclosed embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A vertical semiconductor device comprising: a substrate; a drift region over the substrate, wherein the substrate and the drift region are doped with a dopant of a first type; a plurality of channel regions extending from an upper portion of the drift region, wherein there are no regrowth interfaces between the drift region and the plurality of channel regions; a first material doped with a dopant of a second type and filling trenches between each one of the plurality of channel regions, the second type opposite in polarity to the first type; at least one first contact over at least one of the plurality of channel regions; and at least one second contact over a bottom surface of the substrate.
2. The vertical semiconductor device of claim 1 wherein the drift region is silicon carbide.
3. The vertical semiconductor device of claim 2 wherein the first material is silicon carbide.
4. The vertical semiconductor device of claim 2 wherein the first material is silicon.
5. The vertical semiconductor device of claim 2 wherein the first material is a metal.
6. The vertical semiconductor device of claim 1 further comprising a first layer of the first material that fills a plurality of trenches, wherein the vertical semiconductor device is a transistor, the first material that fills the plurality of trenches forms a plurality of gate regions, the at least one first contact is a source contact, and the at least one second contact is a drain contact; and further comprising at least one gate contact over at least a portion of the first layer.
7. The vertical semiconductor device of claim 6 wherein the at least one gate contact is not provided over the plurality of gate regions.
8. The vertical semiconductor device of claim 6 wherein the at least one gate contact comprises a plurality of gate contacts that are each provided over a corresponding one of the plurality of gate regions.
9. The vertical semiconductor device of claim 1 wherein the drift region and the plurality of channel regions are continuously grown such that there are no regrowth interfaces between the drift region and the plurality of channel regions.
10. The vertical semiconductor device of claim 1 wherein the dopant of the first type is an N-type dopant, and the dopant of the second type is a P-type dopant.
11. The vertical semiconductor device of claim 1 wherein each of the plurality of mesas is an elongated stripe, such that one of a plurality of trenches is provided between adjacent pairs of the plurality of mesas.
12. The vertical semiconductor device of claim 1 wherein each of the plurality of channel regions is

a pillar.

13. The vertical semiconductor device of claim 12 wherein each of the pillars have a circular horizontal cross-section.

14. The vertical semiconductor device of claim 13 wherein each of the pillars has a square horizontal cross-section.

15. The vertical semiconductor device of claim 14 wherein each of the pillars has a polygonal horizontal cross-section.

16. The vertical semiconductor device of claim 1 wherein the drift region comprises a plurality of regions with different doping concentrations.

17. The vertical semiconductor device of claim 1 wherein each of the plurality of channel regions has a plurality of regions with different doping concentrations.

18. The vertical semiconductor device of claim 1 wherein at least one of the plurality of channel regions or the drift region has at least one region with a graded doping profile.

19. The vertical semiconductor device of claim 1 wherein the drift region comprises at least one charge spreading layer below the plurality of channel regions, and at least one drift layer.

20. The vertical semiconductor device of claim 1 wherein the vertical semiconductor device is a diode.

21. The vertical semiconductor device of claim 1 wherein a doping concentration of the second type for the first material is at least two times that of a doping concentration of the first type in the plurality of channel regions.

22. The vertical semiconductor device of claim 1 wherein a width at a most narrow portion of each of the plurality of channel regions is between one and two times that of a height of each of the plurality of channel regions.

23. The vertical semiconductor device of claim 1 wherein the substrate and the drift region comprise a first material system.
